INDIVIDUAL CONTRIBUTOR DOSSIER • AI DEVELOPMENT & CORE ENGINEERING LEAD

Dhruv Patil

Team: Reconstruct.ai | **Institution:** JSPM RSCOE, Pune

Event: We Code Coders Hackathon

Domain: AI Pipelines, LLM Orchestration, Epistemic NLI Verification & Rapid AI Coding

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Repository: github.com/Dhruvplaz-web/reconstruct-ai

1. Personal Mandate & Individual Contribution Statement (Rule 14)

As the AI Development Lead, I own the machine intelligence, reasoning, and automated code implementation engine of Reconstruct.ai. I design the multi-stage LLM orchestration pipelines (leveraging Gemini 1.5 Pro and Claude 3.5 Sonnet) that autonomously extract architectural decisions, unresolved blockers, and operational dependencies from raw Git commit histories and unstructured chat threads. Furthermore, I engineered the mathematical formulation for Epistemic Confidence Scoring G(A) and the pairwise Natural Language Inference (NLI) Contradiction Radar using HuggingFace DeBERTa-v3 to resolve conflicting project assertions without ungrounded hallucinations.

2. Subsystem Ownership & Technical Architecture

- **Dual-Stage LLM Extraction Chains:** High-recall structured prompt architecture converting PR reviews, commits, and Slack threads into temporal facts.
- **Epistemic Contradiction Radar:** Pairwise NLI cross-encoder (DeBERTa-v3-small) identifying cross-source factual conflicts with >70% contradiction probability.
- **Mathematical Grounding Score Formula:** Implementation of $G(A) = \lambda \cdot S_{evidence} + (1-\lambda) \cdot e^{(-\gamma \Delta t)} \cdot (1-P(\text{Contradiction}))$ to quarantine uncertain assertions.
- **AI-Augmented Core Codebase Development:** Driving rapid full-stack scaffolding and algorithmic logic using state-of-the-art agentic engineering workflows.

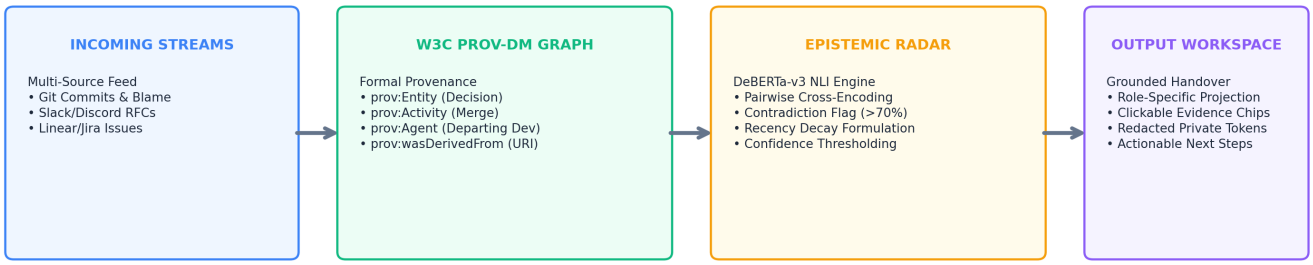


Figure 1: W3C PROV-DM & DeBERTa-v3 NLI Verification Engine Engineered by Dhruv Patil

3. Technical Deliverables Across All 3 Rounds

Hackathon Round	Personal Responsibilities & Concrete Engineering Deliverables
Round 1: The Plan (Current Phase)	Formulated the epistemic verification architecture, prompt schemas, mathematical confidence decay models, and engineered the synthetic benchmark evaluation dataset (Acme Payments microservice).
Round 2: The Build (Working Functional MVP)	Implement the core LLM decision extraction pipeline, citation grounding linker, and local DeBERTa-v3 NLI cross-encoder scripts integrated into the FastAPI backend.
Round 3: The Finish (Production Polish & UAT)	Deliver real-time Contradiction Radar badge triggers in the UI, fine-tune the epistemic confidence thresholds, and deploy automated hallucination prevention guardrails.

4. Competitor Failure Analysis & Defense for this Domain

Existing RAG tools (like NotebookLM or Atlassian Rovo) fail catastrophically when sources contradict each other—they hallucinate an ungrounded compromise. Reconstruct.ai refuses to invent consensus. Our pairwise NLI engine detects contradictions explicitly and surfaces them side-by-side with confidence scores.

5. Judge Viva & Technical Decision Defense (Rule 16 Preparation)

Per Rule 16 ('Judges may ask any team member to explain the product, demonstrate a feature, justify a technical decision or verify their contribution'), this section details the anticipated questions and defensible justifications:

Anticipated Judge Question / Challenge	Defensible Technical Justification & Verification Evidence
Q1: Why use DeBERTa-v3 for contradiction detection instead of asking an LLM directly?	Cross-encoder NLI models like DeBERTa-v3-small are deterministic, computationally lightweight, run locally in milliseconds with zero API costs, and are mathematically trained specifically on Natural Language Inference (premise vs. hypothesis entailment/contradiction).
Q2: How does your architecture solve cross-source entity disambiguation (e.g. Slack chat vs PR commit)?	We implement a 48-hour sliding temporal co-occurrence window combined with semantic embedding cosine similarity. If a Slack thread discussions coincide temporally with a Git PR merge and exhibit cosine similarity > 0.78, we bind them under a formal W3C prov:wasDerivedFrom relationship.
Q3: How do you guarantee that generated handovers contain zero hallucinations?	Through strict W3C PROV-DM cryptographic URI grounding. Every single claim is bound to an exact commit diff or chat thread URI. Assertions with grounding scores G(A) < 0.65 are quarantined into the human review queue and never presented as verified organizational truth.

6. Full Project Team Roster & Peer Verification

Member Full Name	Assigned Role	Institution	Project Verification Status
Khushi Patil	Team Lead & Head of Research	JSPM RSCOE, Pune	**VERIFIED (Team Lead)**
Dhruv Patil	AI Development & Core Engineering Lead	JSPM RSCOE, Pune	**VERIFIED (AI Core Lead)**
Yuvraj Kadam	Lead Full-Stack Frontend & UI/UX	JSPM RSCOE, Pune	**VERIFIED (Frontend Lead)**
Nimisha Joshi	Systems Backend, Ingestion & Security	JSPM RSCOE, Pune	**VERIFIED (Backend Lead)**

Primary Project Repository: https://github.com/Dhruvplaz-web/reconstruct-ai • **Target Event:** We Code Coders Three-Round Hackathon